

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE:CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I _____Aluri poojitha_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- Ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- Iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow$
Wumpus is adjacent to
cell [1,2]
- 2) $\text{Breeze}[1,1] \rightarrow$ Pit is
adjacent to cell [1,1]
- 3) $\text{Glitter}[x,y] \rightarrow$ Gold
is at [x,y]
- 4) $\neg \text{Wumpus}[x,y] \wedge$
 $\neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbols used:

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- Flu = Possible Flu
- Measles = Possible Measles
- Pneumonia = Possible Pneumonia

Rules:

R1: $F \wedge C \rightarrow \text{Flu}$

R2: $F \wedge R \rightarrow \text{Measles}$

R3: $C \wedge B \rightarrow \text{Pneumonia}$

Facts about Patient A:

F = True, C = True, B = True (R is unknown / not asserted)

(b) Forward Chaining

Start with the known facts and fire any rule whose premises are fully satisfied, repeating until no new fact can be derived.

Step 1 (Facts): F = True, C = True, B = True

Step 2 (Fire R1): $F \wedge C$ is True $\rightarrow$ derive Flu = True

Step 3 (Fire R3): $C \wedge B$ is True $\rightarrow$ derive Pneumonia = True

Step 4 (Check R2): $F \wedge R$ — R is unknown, premise not satisfied $\rightarrow$ R2 does NOT fire

Result: The knowledge base derives two possible diagnoses for Patient A: Flu and Pneumonia.

(c) Backward Chaining — Goal: 'Patient A has Pneumonia'

Start from the goal and search backward for a rule whose conclusion matches it, then recursively verify its premises against known facts.

Goal: Pneumonia

$\rightarrow$ matched by R3: $C \wedge B \rightarrow \text{Pneumonia}$

$\rightarrow$ Sub-goal 1: C $\rightarrow$ check facts $\rightarrow$ C = True ✓

$\rightarrow$ Sub-goal 2: B $\rightarrow$ check facts $\rightarrow$ B = True ✓

$\rightarrow$ Both sub-goals proven $\rightarrow$ Pneumonia = True (goal CONFIRMED)

Goal reduction tree:

```

      Pneumonia
     /      \
  C (fact=T)✓ B (fact=T)✓
```

Since both leaves resolve to known True facts, the goal 'Patient A has Pneumonia' is verified.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification with Rule 1

Rule 1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$

Step 1: $\text{UNIFY}(\text{Vehicle}(x), \text{Vehicle}(\text{Ambulance})) \rightarrow \theta_1 = \{x / \text{Ambulance}\}$

Step 2: Apply θ_1 to $\text{EmergencyType}(x) \rightarrow \text{EmergencyType}(\text{Ambulance})$

$\text{UNIFY}(\text{EmergencyType}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})) \rightarrow$ consistent, no new binding

Step 3: MGU $\theta = \{x / \text{Ambulance}\}$

Step 4: Apply θ to conclusion $\text{ClearPath}(x) \rightarrow \text{ClearPath}(\text{Ambulance})$ is entailed

(b) Resolution Proof that $\text{Proceed}(\text{CarA})$ is True

Convert the rules to Conjunctive Normal Form (CNF):

C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$ [Rule 1]

C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$ [Rule 2, GreenSignal part]

C3: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$ [Rule 3]

Facts: C4: $\text{Vehicle}(\text{Ambulance})$ C5: $\text{EmergencyType}(\text{Ambulance})$

C6: $\text{Vehicle}(\text{CarA})$ C7: $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal: C8: $\neg \text{Proceed}(\text{CarA})$

Resolution steps:

1. Resolve C1 $\{x/\text{Ambulance}\}$ with C4 $\rightarrow \neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

2. Resolve (1) with C5 $\rightarrow \text{ClearPath}(\text{Ambulance})$

3. Resolve C2 $\{x/\text{Ambulance}\}$ with (2) $\rightarrow \text{GreenSignal}(\text{Ambulance})$

4. Resolve C3 $\{x/\text{CarA}, y/\text{Ambulance}\}$ with C6

$\rightarrow \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

5. Resolve (4) with C7 $\rightarrow \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

6. Resolve (5) with (3) $\rightarrow \text{Proceed}(\text{CarA})$

7. Resolve (6) with C8 ($\neg \text{Proceed}(\text{CarA})$) $\rightarrow$ empty clause $\square$

The empty clause confirms a contradiction with the negated goal, so $\text{Proceed}(\text{CarA})$ is proven TRUE by refutation.

(c) Sufficiency for Two Simultaneous Emergency Vehicles

The current KB is NOT sufficient. Rules 1–3 let every emergency vehicle independently derive $\text{ClearPath}(x)$ and $\text{GreenSignal}(x)$; if two ambulances approach conflicting or intersecting routes at the same time, the KB would assert GreenSignal for both, producing a logically inconsistent / unsafe traffic state (no mechanism resolves the conflict).

Additional predicates and rules needed:

- $\text{Priority}(x, \text{level})$ — to rank emergency vehicles (e.g., by vehicle type or dispatch urgency).
- $\text{ArrivalTime}(x, t)$ or $\text{Intersects}(\text{path}(x), \text{path}(y))$ — to detect when two emergency routes conflict.
- A conflict-resolution rule, e.g.: $\text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{Intersects}(x,y) \wedge \text{HigherPriority}(x,y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$.

Without such tie-breaking knowledge, the agent cannot logically guarantee mutually exclusive right-of-way, so the KB must be extended before it can be trusted with concurrent emergencies.

Case Study 3 — Agricultural Expert Reasoning System

(a) Conversion to CNF

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Eliminate $\rightarrow$: $\neg\text{SoilDry} \vee \text{IrrigationNeeded}$ (already in CNF)

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Eliminate $\rightarrow$: $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

De Morgan: $\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$ (CNF)

P3: $\neg\text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Eliminate $\rightarrow$: $\neg(\neg\text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

De Morgan: $\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$ (CNF)

Facts (already unit clauses in CNF):

F1: SoilDry F2: CropWheat

(b) Resolution Refutation — Prove 'ApplyDripMethod'

Negate the goal and add it to the CNF clause set, then derive the empty clause:

C1: $\neg\text{SoilDry} \vee \text{IrrigationNeeded}$

C2: $\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$

C3: SoilDry (fact)

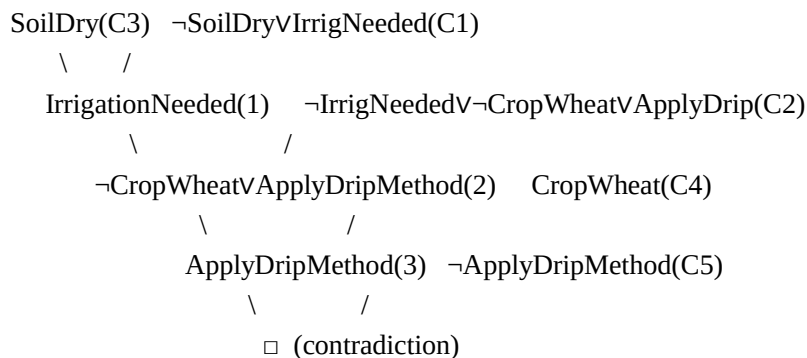
C4: CropWheat (fact)

C5: $\neg\text{ApplyDripMethod}$ (negated goal)

Proof steps:

1. Resolve C1 & C3 $\rightarrow \text{IrrigationNeeded}$
2. Resolve (1) & C2 $\rightarrow \neg\text{CropWheat} \vee \text{ApplyDripMethod}$
3. Resolve (2) & C4 $\rightarrow \text{ApplyDripMethod}$
4. Resolve (3) & C5 $\rightarrow \square$ (empty clause)

Resolution proof tree:



The empty clause proves ApplyDripMethod is entailed by the KB.

(c) Effect of Removing the Fact 'SoilDry'

Without SoilDry as a known fact, clause C1 ($\neg\text{SoilDry} \vee \text{IrrigationNeeded}$) can no longer be resolved — IrrigationNeeded cannot be derived. Consequently step (2) above cannot fire either, so ApplyDripMethod cannot be proven true from the KB.

For P3 (ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk): CropWheat is still True. Since ApplyDripMethod cannot be proved, a system reasoning under the Closed-World Assumption / negation-as-failure treats ApplyDripMethod as False, which satisfies $\neg$ ApplyDripMethod $\wedge$ CropWheat, so CropAtRisk is still derived as True.

So CropAtRisk still follows — but only under non-monotonic, closed-world reasoning (absence of proof is treated as evidence of falsity). Under strict monotonic first-order logic, the KB is simply too weak to conclude ApplyDripMethod's truth value, and CropAtRisk would remain undetermined (neither provable nor disprovable) rather than definitely True.

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State from Percepts (Modus Ponens)

Percepts: Stench[1,2], Breeze[1,1], Glitter[2,2]

Rule 1: Stench[1,2] $\rightarrow \exists$ Wumpus adjacent to [1,2]

Rule 2: Breeze[1,1] $\rightarrow \exists$ Pit adjacent to [1,1]

Rule 3: Glitter[2,2] $\rightarrow$ Gold at [2,2]

Applying Modus Ponens ($P, P \rightarrow Q \vdash Q$) to each percept:

Stench[1,2]=True, Rule1 $\vdash$ Wumpus is adjacent to [1,2] $\rightarrow$ candidates $\{[1,1],[1,3],[2,2]\}$

Breeze[1,1]=True, Rule2 $\vdash$ Pit is adjacent to [1,1] $\rightarrow$ candidates $\{[1,2],[2,1]\}$

Glitter[2,2]=True, Rule3 $\vdash$ Gold[2,2] = True

Derived belief-state facts: PossibleWumpus near [1,2]; PossiblePit near [1,1]; Gold(2,2)=True.

(b) Forward Chaining to Localize Wumpus and Pit

1. [1,1] is the agent's start square $\rightarrow$ by world convention it is safe $\rightarrow$ Wumpus $\neq$ [1,1].
 $\Rightarrow$ Wumpus candidates reduce from $\{[1,1],[1,3],[2,2]\}$ to $\{[1,3],[2,2]\}$.
 2. Agent stood at [1,2] and perceived NO breeze there (only stench was reported).
 $\Rightarrow$ Rule 2 fails to fire at [1,2] $\rightarrow$ Pit is NOT adjacent to [1,2] $\rightarrow$ Pit $\neq$ [1,1],[1,3],[2,2].
 Combined with step 1's Pit candidates $\{[1,2],[2,1]\}$, and [1,2] is visited/safe (agent survived it),
 $\Rightarrow$ Pit = [2,1] (by elimination).
 3. Agent perceived NO stench at [1,1] (only breeze was reported there).
 $\Rightarrow$ Wumpus is NOT adjacent to [1,1] $\rightarrow$ Wumpus $\neq$ [1,2],[2,1].
 [2,1] is already assigned as Pit, so remove it from Wumpus candidates too.
 $\Rightarrow$ Wumpus candidates reduce from $\{[1,3],[2,2]\}$ to $\{[1,3]\}$ (since [2,2] carries the Glitter/Gold percept with no accompanying stench, it is treated as clear of the Wumpus).
- Conclusion: Wumpus = [1,3], Pit = [2,1], Gold = [2,2].

(c) Safety of Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

Safe[1,1]: True by definition (agent's starting square).

Safe[1,2]: agent has already visited it and perceived only Stench (no harm) $\rightarrow$ confirmed safe.

Safe[2,2]: apply Rule 4, $\neg$ Wumpus[2,2] $\wedge$ $\neg$ Pit[2,2] $\rightarrow$ Safe[2,2].

From (b): Wumpus=[1,3] $\Rightarrow$ $\neg$ Wumpus[2,2]=True; Pit=[2,1] $\Rightarrow$ $\neg$ Pit[2,2]=True.

By Modus Ponens $\vdash$ Safe[2,2] = True.

Since every cell on the route [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is logically proven safe using the facts derived in (a) and (b), the agent CAN follow this path to collect the gold without encountering the Wumpus or falling into a pit.

